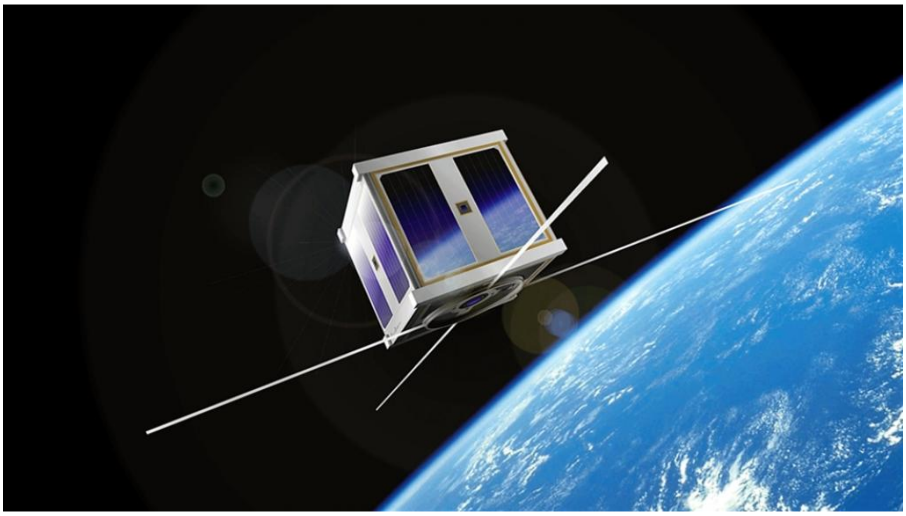
Project Report

# The Art of Scientific Computing: CubeSat Thermal Modelling

*Author:*
Yi Shuen Lee (796840)

*Subject Handbook code:*
COMP90072



Faculty of Science
The University of Melbourne

31st May 2019

**Project Supervisors:** A/Prof. Roger Rassool & A/Prof. Michele Trenti

# Contents

# Abstract

An important aspect to be considered when designing a spacecraft is its thermal properties. An overheated or overcooled satellite in space may potentially lose its optimised functionality and in extreme cases, some components of the spacecraft may also be damaged. Thermal modelling of a practical satellite is extremely complex and can be computationally expensive. Thus, this project aims to investigate the temperature distribution across the infinitesimally thin 2-dimensional (2-D) surface of a simplified hollow 1U CubeSat made out of pure homogeneous aluminium. The temperature distribution of the Cubesat is essentially the solution to the heat equation of six 2-D planes which represents the six infinitesimally thin surfaces of the CubeSat. The heat equation is first approximated to a finite difference equation using Finite Difference Method (FDM) before it is solved using the Explicit Euler's method. Due to stability criteria that arises from numerical techniques, the heat flow simulation of the CubeSat is limited to only a few seconds. The thermal model developed in this project is also based on many assumptions and a highly simplified CubeSat compared to practical situations. However, it has provided valuable insights to the thermal evolution of a CubeSat under the given constraints. The simulation successfully demonstrated the isotropic distribution of heat in a homogeneous material and direction of heat flow down the temperature gradient. With those two crucial properties of heat flow obeyed, the simulation is deemed physically valid. The simulation also showed that the overall temperature of the CubeSat has doubled within $5s$ of placing the CubeSat in orbit. This implies that if the CubeSat was left in orbit for a long time, the temperature will increase endlessly as the radiation of energy outwards via Stefan-Boltmann Law is not fast enough to expel the incoming heat from the Sun and the Earth. This result is the main takeaway from this project.

# Introduction

## Background

CubeSats are small, standardised satellites that can be classified based on the criteria they satisfy. Unlike regular satellites, the dimension (size) and shape of CubeSats are universally fixed so that they can be categorised is based on these properties. CubeSats come in multiples of a standard, customary unit ‘U’ where 1U connotes the dimension $10cm \times 10cm \times 10cm$. Thus, CubeSats are sometimes also referred to as “U-class spacecrafts”. Common CubeSat classes are 1U, 2U, 3U and 6U, where a 1U CubeSat typically weighs between 1 to 1.33kg. [9]

Although the dimension and shape of CubeSats are specified, there are many other aspects that has to be taken into consideration when designing the instrument. An ideal CubeSat is optimised such that it maximally reduces the building and transporting costs. A few examples of facets that play a crucial role when designing a CubeSat include the material, hardware components, thermal control, functionality and efficiency.

## Project Report Overview

This project aims to investigate the thermal properties of a 1U CubeSat by developing a numerical model that simulates heat flow in a CubeSat. Assuming that the entire CubeSat is made of the same homogeneous material, the homogeneous heat equation is then a good point to start.

The next step is to think about how to approach the problem using the homogeneous heat equation. It turns out that finite difference method (FDM) is an appropriate option since the CubeSat is made up of 6 square faces as the system satisfies the restrictions for the use of FDM. This technique approximates the heat equation, a second order differential equation, into a finite difference equation. The discretised heat equation can then be solved using the Explicit (Forward) Euler Method where the resulting solution is a function that describes the time evolution of temperature at any given point in space of the relevant system. The key concepts and procedures for these numerical methods are elaborated in Section 1 along with examples of their applications on a 1-D strip and a 2-D plane.

With these practical numerical tools in hand, one can start to think about how to apply them to the CubeSat thermal modelling problem. It would definitely be ideal to come up with an accurate model for the heat flow within a practical CubeSat, but unfortunately, the complexity of this problem increases rapidly with increasing physical precision. For that reason, constraints have to be set out to initialise this difficult problem. This project explores diffusion of heat across the 2-dimensional surface of a highly simplified CubeSat. Section 2 narrates the specifications of this simplified CubeSat and establishes assumptions that are to be made for the development of the CubeSat thermal model.

Section 3 then outlines of the algorithm and functionalities of the model developed under the given constraints. The results of the simulation as well as its interpretations are subsequently presented in the Section 4. The final section then concludes the project and its findings, and discusses some measures that can be taken to improve the practicality of this model so that it becomes more applicable to practical situations.

# Section 1

# Finite Difference Method (FDM)

This is an introductory section which will outline the methodologies of approximating the heat equation using the finite differente method (FDM). It begins with an introduction to the homogeneous heat equation equation and its significance in the context of CubeSat thermal modelling, followed by a discussion on the concept of finite difference method (FDM) which can be employed to discretise the continuous heat equation. Once the heat equation is discretised, it can solved using the Explicit Euler Method. The procedures and applications of this numerical technique will be demonstrated through two simple examples: a 1-D strip and a 2-D Plane.

## 1.1 The Homogeneous Heat Equation

The homogeneous heat equation is a second order differential equation which describes the isotropic flow of heat in a homogeneous medium. In absence of a heat source or heat sink, the heat equation in any dimension (1-D, 2-D or 3-D) takes the form of a Laplace equation

$$\frac{\partial u}{\partial t} = \alpha \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \tag{1.1}$$

where $u = u(x, y, z, t)$ is temperature and $\alpha$ is thermal diffusivity of the medium in units of length squared per unit time.

However in most practical situations, the considered system is not isolated. This means that there must be an extra term added to the heat equation called the temperature flux, $du/dt$ which accounts for the net increase or decrease in temperature of the whole system due to an external heat source or sink. In the case of a CubeSat, one would expect heat input from sources the Sun and the Earth resulting in an overall increase in the CubeSat temperature. Furthermore, the CubeSat itselt will also radiate excessive heat into space causing an overall temperature to decrease. The sum of these two temperature changes gives the net temperature flux of the CubeSat.

However, it is actually much easier to determine the net heat energy flux into or out of the system $dQ/dt$ than it is to determine the net temperature flux $du/dt$. It would then be convenient to find the relationship between these two quantities, starting from the heat capacity equation

$$\Delta Q = m c_p \Delta u, \tag{1.2}$$

where $\Delta Q$ is the change in heat energy, $\Delta u$ is the change in temperature and $c_p$ is the specific heat capacity of the given uniform medium. Then divide both sides of Equation 1.2 by a finite change in time, $\Delta t$ and the volume of the medium, $V$ to get

$$\frac{\Delta Q}{V \Delta t} = m c_p \frac{\Delta u}{V \Delta t}, \tag{1.3}$$

This equation can then be taken to the infinitesimal limit to obtain

$$\begin{aligned}\frac{dQ}{Vdt} &= \rho c_p \frac{du}{dt} \\ \frac{du}{dt} &= \frac{1}{\rho c_p}\frac{dq}{dt}\end{aligned} \tag{1.4}$$

where $q$ has been defined to be the heat energy per unit volume anc $\rho = {}^m/_V$ is the density of the medium.

Equation 1.4 is precisely the net temperature influx term expressed in terms of heat flux per unit volume, which can be added as a new term into Equation 1.1 in the presence of a heat source or sink. The new heat equation with the additional flux term is then no longer a Laplace equation, but rather a Poisson equation

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) + \frac{1}{\rho c_p}\frac{dq}{dt}. \tag{1.5}$$

Noting that the thermal diffusivity $\alpha$ is by definition

$$\alpha = \frac{k}{\rho c_p} \tag{1.6}$$

where $k$ = thermal conductivity in units of power per unit length per unit temperature, $\rho$ = density and $c_p$ = specific heat capacity of the medium in which $u$ propagates [10], Equation 1.5 can be re-expressed as

$$\frac{\partial u}{\partial t} = \alpha\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} + \frac{1}{k}\frac{dq}{dt}\right). \tag{1.7}$$

## 1.2 Finite Difference Approximation of the Heat Equation

Finite difference method (FDM) is a commonly used numerical method for solving physical problems that involve partial differential equations. FDM approximates a continuous, partial differential equation to a discretised, finite difference equation so that it can be solved using other numerical techniques. FDM is not to be confused with finite element method (FEM), which will not be used in this context. While the FEM can be used to solve systems with complex geometry, while the FDM only approximates the differential equation that describes the given system. FDM is also restricted to systems that can be divided up into regularly shaped elements like 1-D strips, 2-D square or rectangular planes, 3-D cuboids etc. [6] Since the CubeSat satisfies the conditions for FDM, this method will be implemented to discretise the heat equation which will be used later to model heat flow on the surface of the CubeSat.

In order to discretise the heat equation, the first step is to perform Taylor series expansion on each of the spatial and temporal parameters of the temperature function $u(x, y, z, t)$. As an example, the Taylor expansion of the parameter $x$ of $u(x', y, z, t)$ with $x' > x$ is

$$u(x', y, z, t) = u(x, y, z, t) + \Delta x \frac{\partial u(x, y, z, t)}{\partial dx} + \frac{\Delta x^2}{2}\frac{\partial^2 u(x, y, z, t)}{\partial x^2} + \mathcal{O}(\Delta x^3) \tag{1.8}$$

where $\Delta x = x' - x$. Then by letting $\Delta x$ be a small but finite number, one can neglect terms of order $\Delta x^2$ and above in Equation 1.8 to obtain

$$u(x', y, z, t) \approx u(x, y, z, t) + \Delta x \frac{\partial u(x, y, z, t)}{\partial dx} \tag{1.9}$$

which can be rearranged to give

$$\frac{\partial u(x, y, z, t)}{\partial x} \approx \frac{u(x', y, z, t) - u(x, y, z, t)}{\Delta x} \tag{1.10}$$

This right hand side of this equation is an approximation to the first order differentiation of $u(x, y, t, z)$ with respect to $x$, which fundamentally describes the forward difference in spatial coordinate $x$. The word ‘forward’ is used here because $x' > x$ as previously stated.

However, the heat equation is a second order partial differential equation so it necessary to perform approximations up to second order differentiation using the FDM. Using $x$ as the differentiated parameter again, take the Taylor series expansion of $u(x'', y, z, t)$, but this time with $x'' < x$

$$u(x'', y, z, t) = u(x, y, z, t) + \Delta x'' \frac{\partial u(x, y, z, t)}{\partial x} + \frac{\Delta x''^2}{2} \frac{\partial^2 u(x, y, z, t)}{\partial x^2} + \mathcal{O}(\Delta x^3) \tag{1.11}$$

where $\Delta x'' = x'' - x$. This is the exact opposite case to 1.10 since $\Delta x'' < 0$ and it is known as the backward difference in spatial coordinate $x$.

In the case of finding a finite difference approximation to the second order differentiation, terms of $\mathcal{O}(\Delta x^3)$ and above are neglected. The $\mathcal{O}(\Delta x^2)$ terms has to be retained, otherwise the second order differential term vanishes. Then by letting $|\Delta x| = |\Delta x''|$ (i.e. same intervals between $x''$,$x$ and $x'$), Equations 1.8 and 1.11 can be combined by addition to obtain the approximation for second order differentiation of $u(x, y, z, t)$ with respect to $x$ [6]

$$\begin{aligned} u(x'', y, z, t) + u(x', y, z, t) &= 2u(x, y, z, t) + \Delta x^2 \frac{\partial^2 u(x, y, z, t)}{\partial x^2} \\ \frac{\partial^2 u(x, y, z, t)}{\partial x^2} &= \frac{u(x'', y, z, t) - 2u(x, y, z, t) + u(x', y, z, t)}{\Delta x^2}. \end{aligned} \tag{1.12}$$

This is known as the central difference in the $x$ spatial coordinate because it involves both the forward and backward steps of $x$.

Given that the differences between $x$, $x'$ and $x''$ are small but finite, the right hand sides of Equations 1.10 and 1.12 are exactly the finite difference approximation for first and second order differentiation respectively. This approximation discretises the spatial coordinate $x$ by dividing it up into finite-sized segments with equal intervals, rather than having a continuum with infinitesimal intervals $dx$.

The whole procedure can be repeated in the exact same way for $y$ and $z$, and up to the first order for $t$, to obtain the spatially and temporally discretised Heat Equation.

## 1.3 Explicit Euler Method

The Explicit (Forward) Euler Method numerical technique used to solve initial value problems that evolves forward in time, where an initial value problem is a differential equation with initial conditions provided. This method is sometimes also known as the Forward Time Central Space (FTCS) Method. [6]

Numerical methods like the Explicit Euler Method typically involve iterations and are usually applicable to continuous differential equation with infinite iterations due to continuity. However, a numerical loop cannot go on infinitely, so it is essential to discretise these differential equations beforehand.

Given an initial condition, the Explicit Euler Method can then be employed to solve the heat equation which has been discretised in Secion 1.2. The solution to the heat equation reveals the time evolution of temperature at each point in space.

However, due to the truncation of the Taylor Series up to a given order in the finite difference approximation of the heat equation, there will be in increasing error with every increasing temporal or spatial step. This introduces a constraint within which the approximation would converge to the exact solution and this constraint is known as the "stability criteria". [8] The stability criteria varies for different dimensions of the heat equation, or any other differential equations, although they are analogous. Stability criteria of the 1-D and 2-D heat equations will be discussed and shown in the proceeding subsections as they are relevant in the context of the CubeSat modelling problem.

## 1.4 Applications of the Explicit Euler Method

### 1.4.1 Heat Diffusion on a 1-D Strip

This section investigates the heat diffusion across a 1-D strip. To carry out finite difference approximation, the full length $L$ of the strip will be divided into $N$ segments, each of which has length of $\Delta x$ and the timespan $T$ across which diffusion is investigated will be divided into $t$ time steps, each of has a duration of $\Delta t$ . One can then denote the temperature in the $i$-th segment at the $n$-th time step as $u_i^n$, with $i$ and $n$ starting from the zeroth index.

The specifications for this example is given by: $L = 10cm$, $N = 10$, $\Delta x = 1cm$, $T = 50s$, $t = 1000$, $\Delta t = 0.01s$ and is shown in the figure below

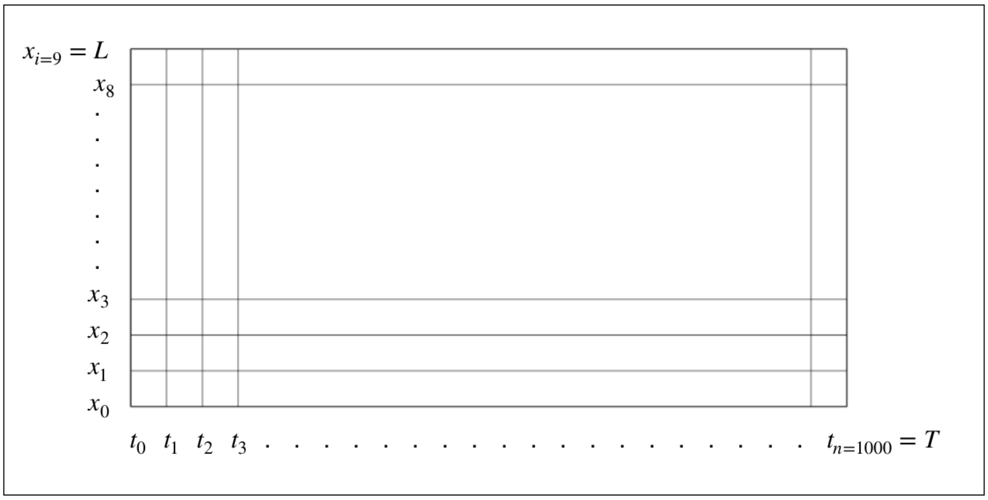



**Figure 1.1:** Visualisation of the discretisation in both the spatial and temporal dimensions.[6]

Given that the 1-D strip has only one spatial dimension and assuming that there are no heat source or sink for the moment, the following 1-D heat equation applies

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}. \tag{1.13}$$

Discretising this equation using the finite difference method as shown in Section 1.2 then results in an equation that looks like this

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \alpha \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}. \tag{1.14}$$

where the new notation $u_i^n$ has replaced the $x$, $x'$ and $x''$ notations. Also, assume that the strip is made of pure aluminium so that $\alpha = 0.97cm^2s^{-1}$.[4] Rearranging Equation 1.14 then gives the temperature evolution of the $i$-th element from time step $n$ to $n+1$

$$u_i^{n+1} = s(u_{i+1}^n + u_{i-1}^n) + (1-2s)u_i^n \tag{1.15}$$

where

$$s = \frac{\alpha \Delta t}{\Delta x^2} > 0. \tag{1.16}$$

Notice that Equation 1.15 will be iterated over many time steps and in each step of iteration, the parameter $s$ is involved. Here is where stability criteria is imposed. In order to ensure a convergent iteration, two conditions where $s < 1$ in the first term and $s \leq 1/2$ in the second term must be satisfied. The overall stability criteria in the 1-D case is then given by$\Delta t \leq \frac{\Delta x^2}{2\alpha}$.

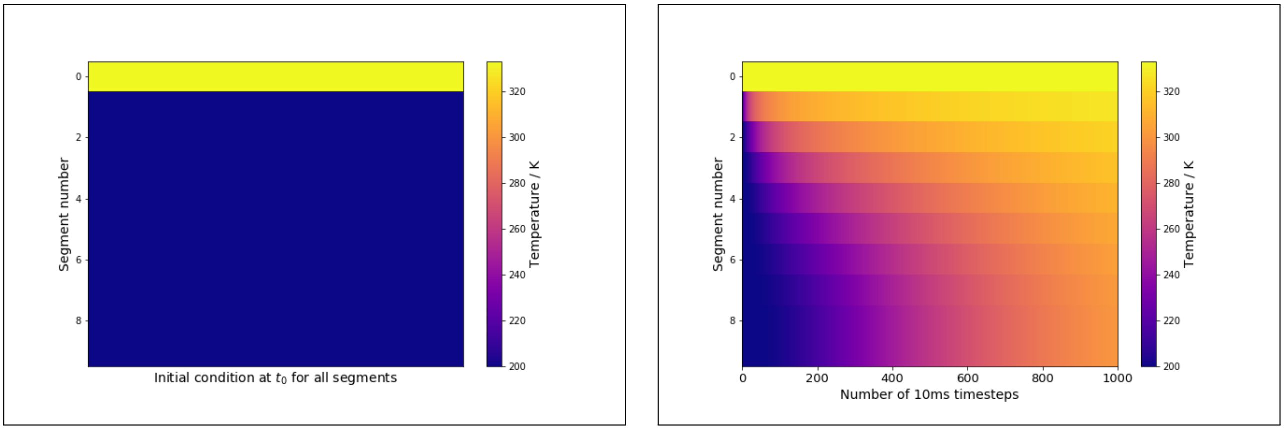
A matrix can now be used to represent the temperature evolution for all segments except for the first and last element since they are at the edge where the central difference cannot be applied. The matrix follows python where indexing starts from 0, so it takes the following form

$$\begin{pmatrix} u_1^{n+1} \\ u_2^{n+1} \\ u_3^{n+1} \\ \vdots \\ u_8^{n+1} \end{pmatrix} = \begin{pmatrix} 1-2s & s & 0 & 0 & 0 & \cdots & 0 \\ s & 1-2s & s & 0 & 0 & \cdots & 0 \\ 0 & s & 1-2s & s & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & \cdots & \cdots & \cdots & s \end{pmatrix} \begin{pmatrix} u_1^n \\ u_2^n \\ u_3^n \\ \vdots \\ u_8^n \end{pmatrix} + s \begin{pmatrix} u_0^n \\ 0 \\ 0 \\ \vdots \\ u_9^n \end{pmatrix} \tag{1.17}$$

where the second term on the right hand side is the boundary condition term.

Boundary conditions must provided for the segments at each end of the strip since their evolution over time cannot be computed using the central difference algorithm. For this example, the first segment will be set to a fixed temperature of $u_0^n = 333K$ for all time steps $n$, implying that there is some form of heat source. By doing this, the temperature flux $\frac{\Delta u}{\Delta t}$ due to an external source has been automatically incorporated into the heat equation although it has not been explicitly added to equation 1.14. For the last segment, there are two options of boundary conditions: one is to set it to its initial temperature for all times (i.e. strip is in contact with a heat sink) or it can be set to be equal to the temperature of the the second last segment (i.e let $u_9^n = u_8^n$) . The latter boundary condition is to allow the temperature of the strip rise without limit under the continuous heat input in the first segment. For this example, the latter condition is chosen as it is more relevant in the context of the CubeSat problem where continuous heat sources like the Earth and the Sun are present.

By setting the initial temperature, $u_i^0$ of all the central segments ($1 < i < 8$) at $200K$ and allowing it to diffuse for $T = 50s$ ( 1000 time steps), one should expect the final temperature distribution as shown in Figure 1.2b with an increase in mean temperature of the whole strip due to consistent heat input with no heat output. The initial condition at $T = 0s$ ($n = 0$) is shown in Figure 1.2a.




**(a)** Initial condition of the strip with mean temperature $T_{mean} = 213.945K$.

**(b)** Temperature distribution in the strip after 50 seconds with mean temperature $T_{mean} = 312.334K$.

**Figure 1.2:** Time evolution of temperature distribution across the 1D strip.

### 1.4.2 Heat Diffusion on a 2-D Plane

Heat diffusion on a 2-D plane is very similar to the 1-D strip case. The few major differences are the stability criteria, the boundary conditions along the edges of the plane and that there are four neighbouring elements (left, right, top, bottom) for each central elements instead of just two (left and right) like the 1-D case.

The 2-D plane in this example is a pure aluminium square sheet with infinitesimal thickness and sides of length $L_x = L_y = 10cm$. For convenience, each of the spatial coordinate will be divided into 100 finite segments, so that $\Delta x = \Delta y = 0.1cm$ and the plane becomes an array of $100 \times 100$ elements. The illustration of this is analogous to Figure 1.1 except that the temporal axis is now replaced with the spatial coordinate $y$ so that the visualisation shows only the spatial dimensions without the temporal dimension. Also, instead of specifying the time step as before, the duration of each time step is provided instead where $\Delta t = 0.001s$ (or $1ms$) and then user can specify the time step over which they wish to investigate the heat distribution of the system. By analogy to the 1-D case, the temperature of a segment at the $i$-th x-coordinate, $j$-th y-coordinate and the $n$-th time step can be denoted by $u^n_{i,j}$.

Since this is a 2-D system, the 2-D heat equation will be used. Again, the temperature flux term can be added in later if required, but for now it is not included so the heat equation is written as

$$\frac{\partial u}{\partial t} = \alpha \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right). \tag{1.18}$$

Using the $u^n_{i,j}$ notation, the finite difference approximation of this equation is then given by

$$\begin{aligned} \frac{u^{n+1}_{i,j} - u^n_{i,j}}{\Delta t} &= \alpha \left( \frac{u^n_{i+1,j} - 2u^n_{i,j} + u^n_{i-1,j}}{\Delta x^2} + \frac{u^n_{i,j+1} - 2u^n_{i,j} + u^n_{i,j-1}}{\Delta y^2} \right) \\ u^{n+1}_{i,j} &= u^n_{i,j} + \frac{\alpha \Delta t}{\Delta x^2}(u^n_{i+1,j} + u^n_{i,j+1} + u^n_{i,j-1} + u^n_{i-1,j} - 4u^n_{i,j}). \end{aligned} \tag{1.19}$$

With the same explanation as the 1-D strip case, the stability criteria here is

$$\begin{aligned} \frac{\alpha \Delta t}{\Delta x^2} &\leq 1 \\ \Delta t &\leq \frac{\Delta x^2}{\alpha} \end{aligned} \tag{1.20}$$

Although the same boundary conditions apply as per the 1-D strip, where the elements at the edges of the square are set to be the same temperature as the neighbouring central elements, it is much harder to represent the time evolution of each central element as a matrix. This is because the boundary term of a 2-D plane is no longer as simple as two individual segments at each end of a strip, rather the term must take into account elements that make up all four edges of the square. However, it is not difficult to write an algorithm for this purpose in python (see Appendix A).

For a change, the heat injection case will be investigated for the 2-D plane rather than having a consistent heat source like the 1-D strip. Consider the situation where an instantaneous heat injection occurs within a $20 \times 10$ square at the centre of the plane which raises the temperature of the given region up to $T = 333K$. At this point of time, other elements in the plane are all of constant temperature at $T = 200K$. This is the initial condition and it is illustrated in the top left corner of Figure 1.3. After the heat injection, the system is left to evolve in isolation from any heat sources or heat sinks. The distribution of temperature across the 2-D plane at different times after the heat injection is then shown in chronological order in Figure 1.3.

One can infer from Figure 1.3 that the injected heat gets distributed evenly and isotropically across the 2-D plane over time. If the system is left for a sufficiently long period of time, the plane will arrive to an equilibrium state where temperature (or heat) is evenly distributed across all elements. This is a consequence of the homogeneous heat equation.

Another important feature worth mentioning from Figure 1.3 is the mean temperature of the 2-D plane after heat injection. Unlike the 1-D strip where heat is constantly being injected into the system, the 2-D plane is isolated after the instantaneous heat injection. Thus, the heat content of the 2-D plane neither increases nor decreases which results in the same average temperature across the 2-D plane beyond the time of heat injection.

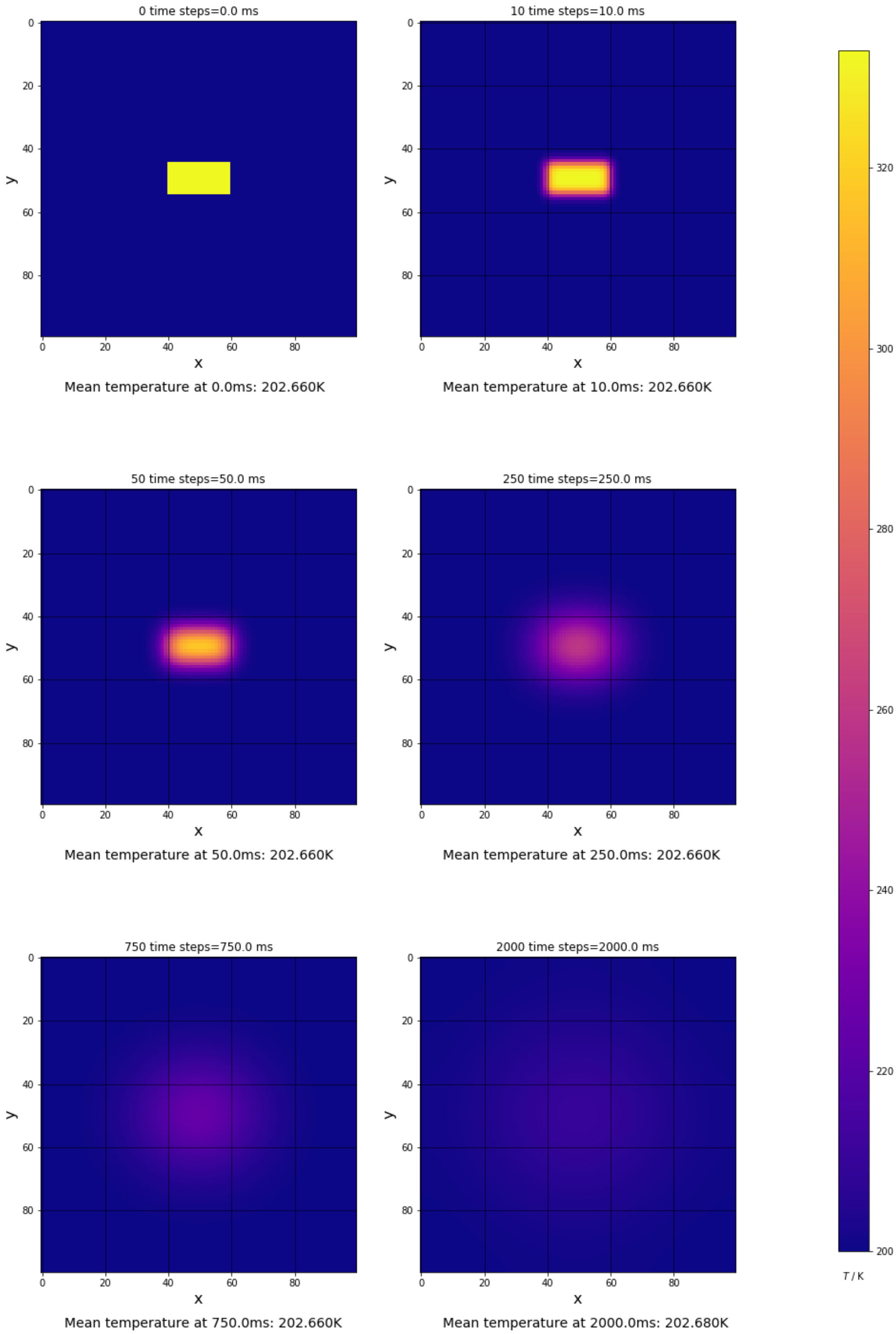



**Figure 1.3:** Temperature distribution across the 2-D plane at a range of time steps after heat injection.

# Section 2

# Modelling Constraints

As previously mentioned, the complexity of a thermal modelling problem increases with increasing physical precision. A good approach to convoluted problems is to simplify it by imposing a range of constraints, and then slowly build upon it as it goes. Fundamentally, the CubeSat itself should not be too complicated. So the first part of this section will elaborate on the features of a simplified CubeSat which the thermal model in this project will be based on. The second half of this section sets down further assumptions that constrains the model.

## 2.1 Specifications of the CubeSat

The model CubeSat investigated in this project will be a hollow, $10cm \times 10cm \times 10cm$ 1U CubeSat made entirely out of pure aluminium. Aluminium is often selected as the main material due to its low density which is good for minimizing the weight of the CubeSat. Since the CubeSat is hollow, there will be no heat flow within the instrument itself. Internal radiation from the heated surfaces of the CubeSat will also be neglected. Hence, heat only flows on the six surfaces of the CubeSat.

A further simplification is to consider all six of the CubeSat faces to be identical and infinitesimally thin, so that the tested 2-D heat flow model constructed in Section 1.4.2 can be applied with minimum modification. The application and modification of the existing 2-D heat flow model will be discussed further in Section 3.

$1U$ CubeSats should also have a standardised mass that lies between $1kg$ to $1.33kg$.[9] Since there are six identical faces on this CubeSat, it would be convenient to set its mass to be $1.2kg$ so that each face of the CubeSat weighs $m_{face} = 0.2kg$.

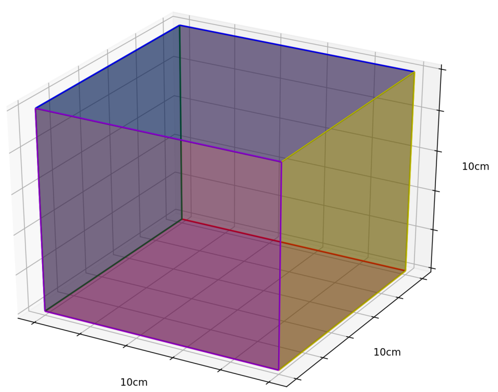



**Figure 2.1:** Visualisation of the hollow, 1U CubeSat with six infinitesimally thin surfaces.

## 2.2 General Assumptions

### 2.2.1 Initial Condition

The initial temperature of the whole CubeSat is set to be at $2.725K$, equivalent to the temperature of the Cosmic Microwave Background.[3] This means the CubeSat have been in deep space for a sufficiently long time that has equilibrated with the background temperature and has just been placed in the Earth's orbit.

### 2.2.2 CubeSat Orbit

In reality, there are plenty of factors that should be taken into account when establishing a practical model for a CubeSat. One of which is the orbital motion of the CubeSat around the Earth and the Sun itself. It would be quite challenging to take this into consideration in the first modelling attempt. To avoid sophistication of the model, the CubeSat is assumed to be constantly $1AU$ from the Sun where orbital motion around Earth is neglected. Also, since most CubeSats have low Earth orbits, it is reasonable to set the distance between the CubeSat and the surface of the Earth to be constant at $600km$.

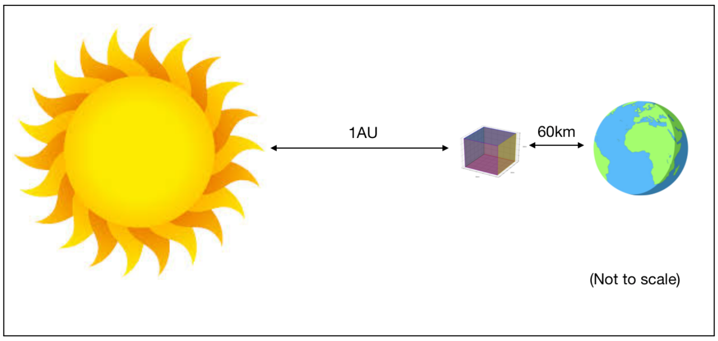



**Figure 2.2:** Visualisation of the simplied CubeSat orbit.

### 2.2.3 CubeSat Heat Absorption

Given the orbital setup as shown in Figure 2.2, the main heat sources that would cause the CubeSat to heat up are the Sun and the Earth. An additional assumption can be made such that only the two surfaces that are directly exposed to the Sun and Earth absorbs the incoming energy, and these surfaces will be labelled Face 1 and Face 3 respectively. Letting Face 1 be the front of the CubeSat, and Face 3 being the back, the remaining four faces: Face 2 (right), Face 4 (left), Face 5 (top) and Face 6 (bottom) are assumed to not have any positive temperature influx. Illustrations of the CubeSat faces can be found in Figure 2.3 on the next page.

Recalling from Equation 1.3 that the temperature flux in 3-D can be calculated from the rate of change of energy density, the analogy in 2-D would be to temperature flux to energy flux $f$, which is the rate of change of energy per unit area. The relationship between temperature and energy flux can be determined the same way using the heat capacity equation. This would result in

$$\begin{aligned} f &= \frac{\Delta Q}{A\Delta T} = mc_p\frac{\Delta u}{A\Delta t} \\ \frac{\Delta u}{\Delta t} &= \frac{A}{mc_p}f \quad \left[\frac{cm^2}{g \times Jg^{-1}K^{-1}}Js^{-1}cm^{-2}\right] \end{aligned} \tag{2.1}$$

where $A$ is the surface area. Dimensional analysis has shown that the right hand side of Equaiton 2.1 is in the correct units, and hence the finite difference 2-D heat equation that incorporates this temperature flux term is

$$u_{i,j}^{n+1} = u_{i,j}^n + \frac{\alpha\Delta t}{\Delta x^2}(u_{i+1,j}^n + u_{i,j+1}^n + u_{i,j-1}^n + u_{i-1,j}^n - 4u_{i,j}^n) + \Delta u \tag{2.2}$$

Notice that the only difference between this equation and Equation 3.1 is the additional $\Delta u$ term.

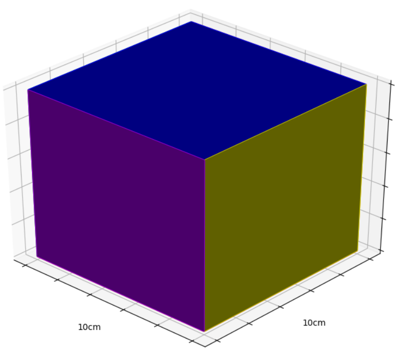
The energy flux from the Sun through Face 1 can be then calculated using the flux-luminosity equation

$$f = \frac{L_\odot}{4\pi d^2} \tag{2.3}$$

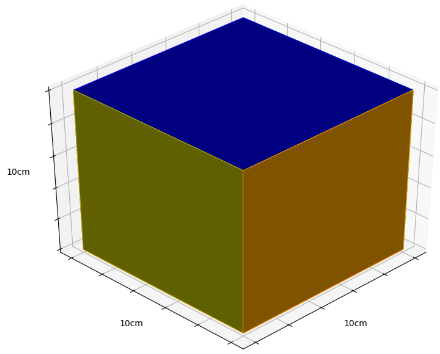
where $L_\odot = 3.828 \times 10^{26} Js^{-1}$ is the intrinsic luminosity of the Sun and $d = 1AU = 1.496 \times 10^{13} cm$ is the distance of the CubeSat from the Sun.[1] The additional term that must be added to the heat equation of Face 1 is then given by

$$\Delta u_{sun} = \alpha_{abs} \frac{A}{mc_p} \frac{L_\odot}{4\pi d^2} \Delta t \tag{2.4}$$

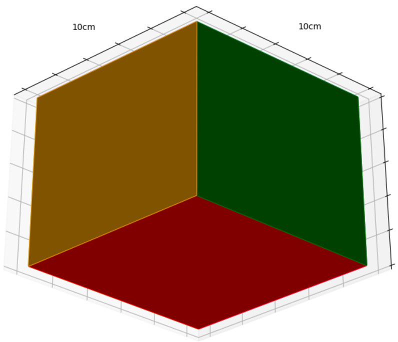
where $A = 100cm^2$ and $m = 200g$. The appropriate values for absorption coefficient and the heat capacity of aluminium are $\alpha_{abs} = 0.04$ and $c_p = 1.01Jg^{-1}K^{-1}$ respectively.[5] [7] The absorption coefficient $\alpha_{abs}$ denotes the fraction of heat that is absorbed by the given surface. It is not to be confused the thermal diffusivity $\alpha$ as they are completely unrelated.



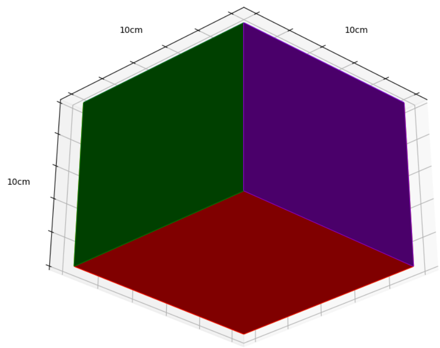

**(a)** Purple = Face 1 (front face, facing the Sun), Yellow = Face 2 (right face), Blue = Face 5 (top face).




**(b)** Yellow = Face 2, Orange= Face 3 (back face, facing the Earth), Blue = Face 5.




**(c)** Orange = Face 3, Green = Face 4 (left face), Red = Face 6 (bottom face).




**(d)** Green = Face 4, Purple = Face 1 (rotated back to the front face), Red = Face 6.

**Figure 2.3:** 4 different orientations of the CubeSat showing all 6 different faces, each with different colour.

The next step is to calculate the energy flux from Earth through Face 3. Since the energy flux due to Earth is the reflected fraction of insolation (i.e. energy flux on Earth due to the Sun), the insolation has to be determined before the calculation can proceed. This is a rather easy process. The first thing is to note that flux is inversely proportional to area

$$f \propto \frac{1}{A} = \frac{1}{4\pi R^2}. \tag{2.5}$$

Then by using Equation 2.3, one can determine the ratio of insolation on the Earth's surface, $f_{earth}$ compared to insolation at the edge of the Sun's surface, $f_{sun}$

$$f_{ratio} = \frac{f_{earth}}{f_{sun}} = \frac{4\pi R_{\odot}^2}{4\pi D^2} \tag{2.6}$$

where $D$ is the distance between the Sun and the Earth (i.e. $1AU = 1.49 \times 10^8 km$) and $R_{\odot}$ is the solar radius valued at $6.957 \times 10^5 km$.[1] In fact, $f_{ratio}$ is also the fraction of flux at the surface of the Sun that reaches the earth. Hence assuming that the Sun is a perfect blackbody with effective temperature $T_{\odot} = 5772K$, $f_{sun}$ can be determined using Stefan-Boltzmann Law

$$f_{sun} = \sigma T_{\odot}^4 \tag{2.7}$$

where $\sigma = 5.67 \times 10^{-8} W m^{-2} K^{-4}$. In consequence, Equation 2.6 can be rearranged to give

$$f_{earth} = \sigma T_{\odot}^4 \frac{R_{\odot}^2}{D^2} \approx 1370 W m^{-2} \tag{2.8}$$

This value can now be used to calculate the solar heat energy reflected upon hitting the Earth's surface.

Given the Earth's radius $R_E$, only $\pi R_E^2$ of the Earth's surface receives the solar insolation, $f_{earth}$ at any given time, yet this energy influx is distributed over the Earth's surface of area $4\pi R_E^2$. The net solar influx through Earth's surface is thus reduced by a factor of 4 where

$$f_{in} = f_{earth} \frac{\pi R_E^2}{4\pi R_E^2} = \frac{1}{4} f_{earth} \tag{2.9}$$

With the Earth's albedo being 0.3 (i.e. the Earth reflects 30% of the incoming energy)[11], the reflected solar flux that affects Face 3 of the CubeSat is

$$f_{out} = 0.3 f_{in} \tag{2.10}$$

and the temperature flux term to be added into the finite different 2-D heat equation of Face 3 is

$$\Delta u_{earth} = \alpha_{abs} \frac{A}{m c_p} f_{out} \Delta t. \tag{2.11}$$

Hence, the overall temperature influx of the CubeSat is

$$\Delta u_{in} = \Delta u_{sun} + \Delta u_{earth} \tag{2.12}$$

### 2.2.4 CubeSat Heat Emission

Revisiting the Stefan-Boltzmann Law, all bodies radiate energy at the rate of

$$f_{body} = \epsilon \sigma T^4 \tag{2.13}$$

where $T$ is the temperature of the body and $\epsilon$ is its emissivity. For a black-body like object, $\epsilon$ is about 1.

This principle implies that the CubeSat must be constantly radiating energy back into space following Equation 2.13. By Kirchoff's Law of Thermal Radiation, the absorptivity and emissivity of an object are equivalent for any given wavelength.[2] Given that $\epsilon = \alpha_{abs}$, the flux emitted by any given element on the CubeSat with $T = u^n_{(i,j)_k}$ can simply be written as

$$f_{out} = \alpha_{abs} \sigma (u^n_{(i,j)_k})^4. \tag{2.14}$$

This flux term is then included in the heat equation of all faces in the form

$$\Delta u_{out} = -\frac{A}{m c_p} \sigma \alpha_{abs} (u^n_{(i,j)_k})^4 \Delta t. \tag{2.15}$$

where the negative sign denotes a negative temperature flux as a result of radiation energy loss.

# Section 3

# CubeSat Thermal Model

With the appropriate numerical tools and the simplified problem in hand, one can now invent a model that describes heat flow on the surface of the CubeSat. This section outlines the steps taken to construct the thermal model

## 3.1 CubeSat Discretisation

Recalling that the simplified CubeSat consists of six indentical faces of infinitesimal thickness, one can replicate the procedures for solving the 2-D heat equation as shown in Section 1.4.2 where each $10cm \times 10cm$ face of the CubeSat is divided up into $100 \times 100$ elements so that $\Delta x = \Delta y = 0.1cm$. It is absolutely crucial to obey stability criteria in order to obtain a convergent solution, so the time step is also set to be $\Delta t = 0.001s$ following the 2-D plane example.

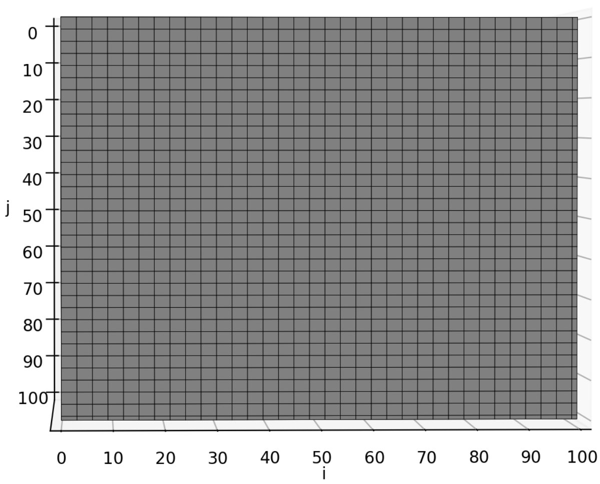



**Figure 3.1:** Visualising a discretised face of the CubeSat. (**Note:** this is not a $100 \times 100$ grid as the individual elements will too small to be seen. This larger grid is shown for the visualisation purposes only.)

Also recall that the temperature for each element in the discretised 2-D plane is denoted by $u^n_{i,j}$. The same can be done here for each face using $u^n_{(i,j)_k}$ where $k$ denotes the face number starting from the zeroth index, so that Face 1 is $k = 0$, Face 2 is $k = 2$ and so on. Hence, the six face of CubeSat can be represented using six individual matrices of size $i \times j = 100 \times 100$, so that the full matrix that represents the whole CubeSat has dimensions of $6 \times 100 \times 100$. When viewed from above (bird's eye view), the $i$ and $j$ labels for all the faces (i.e. the $j$-th row and $i$-th column of each matrix) of the CubeSat follow the orientation shown Figure 3.1. An example of how matrix elements are labelled is described visually in Figure 3.4.

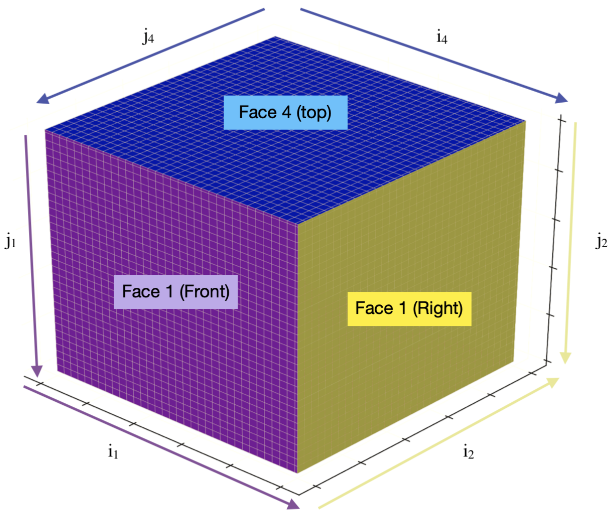



**Figure 3.2:** Discretised version of Figure 2.3a with matrix elements defined for each of the faces. All arrows are pointing from $i/j = 0$ to $i/j = 100$. Subscripts on the $i$'s and $j$'s denote the respective face number.

## 3.2 Heat Flow Modelling Algorithm

The algorithm that solves the 2-D discretised heat equation for all of the six faces of the CubeSat is very similar to that of the single 2-D plane. The central elements of all six faces can be computed using the exact same algorithm given in Appendix A with an additional temperature outflux term $\Delta u_{out}$ for the reasons explained in Section 2.2.3. Also, for Face 1 and Face 3, there will be the temperature influx term from the Sun, $\Delta u_{sun}$ and from the Earth, $\Delta u_{earth}$ respectively.

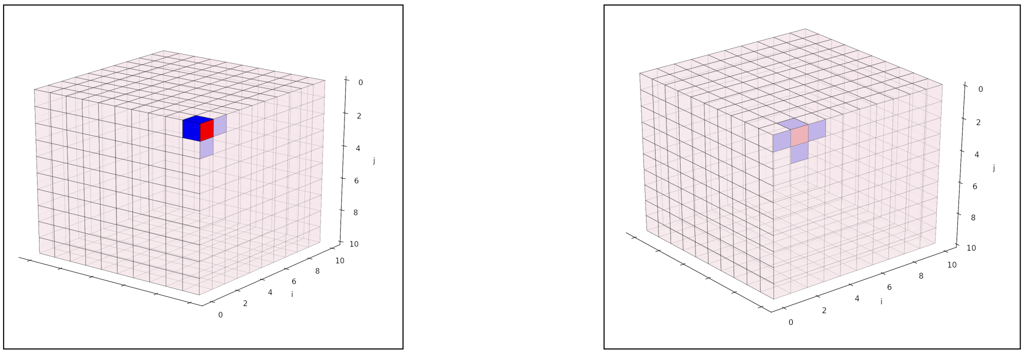
The most distinct difference between the CubeSat and the 2-D plane is that the six faces of the CubeSat are connected at the edge. This implies that the surface of the CubeSat is close and thus there are no boundary conditions. The challenge to generalising the 2-D case for the CubeSat lies in finding the solutions to matrix elements at the four corners and four edges of each face.

To go about this problem, analyse the solution to the 2-D discretised heat equation of the CubeSat where the temperature flux is present

$$\begin{aligned} u^{n+1}_{(i,j)_k} = u^n_{i,j)_k} + \frac{\alpha \Delta t}{\Delta x^2}(u^n_{(i+1,j)_k} + u^n_{(i,j+1)_k} + u^n_{(i,j-1)_k} + u^n_{(i-1,j)_k} - 4u^n_{(i,j)_k}) \\ + \Delta u_{out} + \Delta u_{sun}\delta_{k,0} + \Delta u_{earth}\delta_{k,2} \end{aligned} \tag{3.1}$$

where $\delta_{k,m}$ is the Kronecker delta function. Notice that only the four elements with direct edge-to-edge contact with the concerned element $u^{n+1}_{(i,j)_k}$ directly influences the solution. In that case, some of the neighbours of a corner or an edge element are elements from the neighbouring face. Figure 3.3 illustrates this.

For Face 1 and Face 3 where the edge and corner elements have different temperature flux all other faces, the solutions have to be determined exclusively for the given face. Conversely, the symmetries of the CubeSat allow Faces 2, 4, 5, and 6 to share the same solution, except that the matrix elements have to be readjusted. This means that only one algorithm needs to be written for either one of the faces involved in the symmetry and the rest can be deduced from there. The model developed here writes the algorithm for Face 4 (left). Going by how the matrix elements are defined in Figure 3.1, the solution for Face 4 is then the mirror along the middle column of the Face 2 (right) solution since they are directly opposite one another. The solution for Face 6 (bottom) on the other hand is the transpose of Face 4, and from that, the solution to Face 5 can be deduced by mirroring the matrix of Face 6 along the central column with the same reasoning as to Face 2 being the mirror of Face 3 (see Appendix B for flowchart).



**(a)** Neighbouring elements (blue) of the corner element concerned (red). Two of the neighbouring elements are from two different neighbouring faces.

**(b)** Neighbouring elements (blue) of the edge element concerned. One of the neighbouring elements is from its neighbouring face.

**Figure 3.3**

## 3.3 Model Testing

Before proceeding to simulate the heat flow on the surface of the CubeSat in more realistic conditions, it is always a good measure to test the model first.

To do so, consider the a scenario analogous to the 2-D plane example where there is a instantaneous heat injection such that all the elements in Face 1 rises up to a temperature of $333K$ while the rest of the CubeSat is at $200K$. The heat source is then removed and the CubeSat is left to evolve in isolation. Also assume that there are no heat emission from any part of the CubeSat. This means that all the heat energy should retain in the CubeSat. Applying the model to this scenario yields the results shown as shown below.

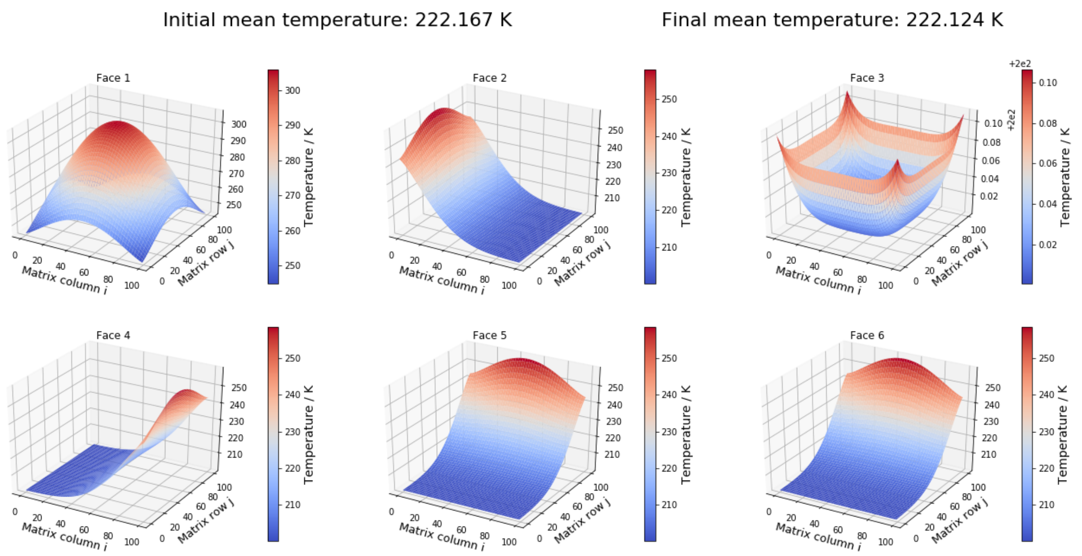



**Figure 3.4:** Temperature distribution on each of the faces after 5 seconds (5000 $1ms$ timesteps).

By symmetry of the CubeSat, Face 2, 4, 5 and 6 should be equivalent at their respective orientations, so the results of the simulation has passed the first confirmation test. The next thing to check for is the symmetric temperature distribution in Faces 1 and 3 since their edges are connected to four symmetric faces. From the results displayed above, this condition is also satisfied. Finally, notice that after 5000 timesteps, the final mean temperature of the system is still about at $\sim 99.98\%$ the initial mean temperature. This implies the model is not entirely perfect because there should not be any reduction in temperature for an isolated system, but is accurate up to a limit. To sum up this test, the viability of the model has been validated and it can now be used to simulate the heat flow on the CubeSat under physical constraints.

# Section 4

# Results and Discussion

## 4.1 Simulation Results

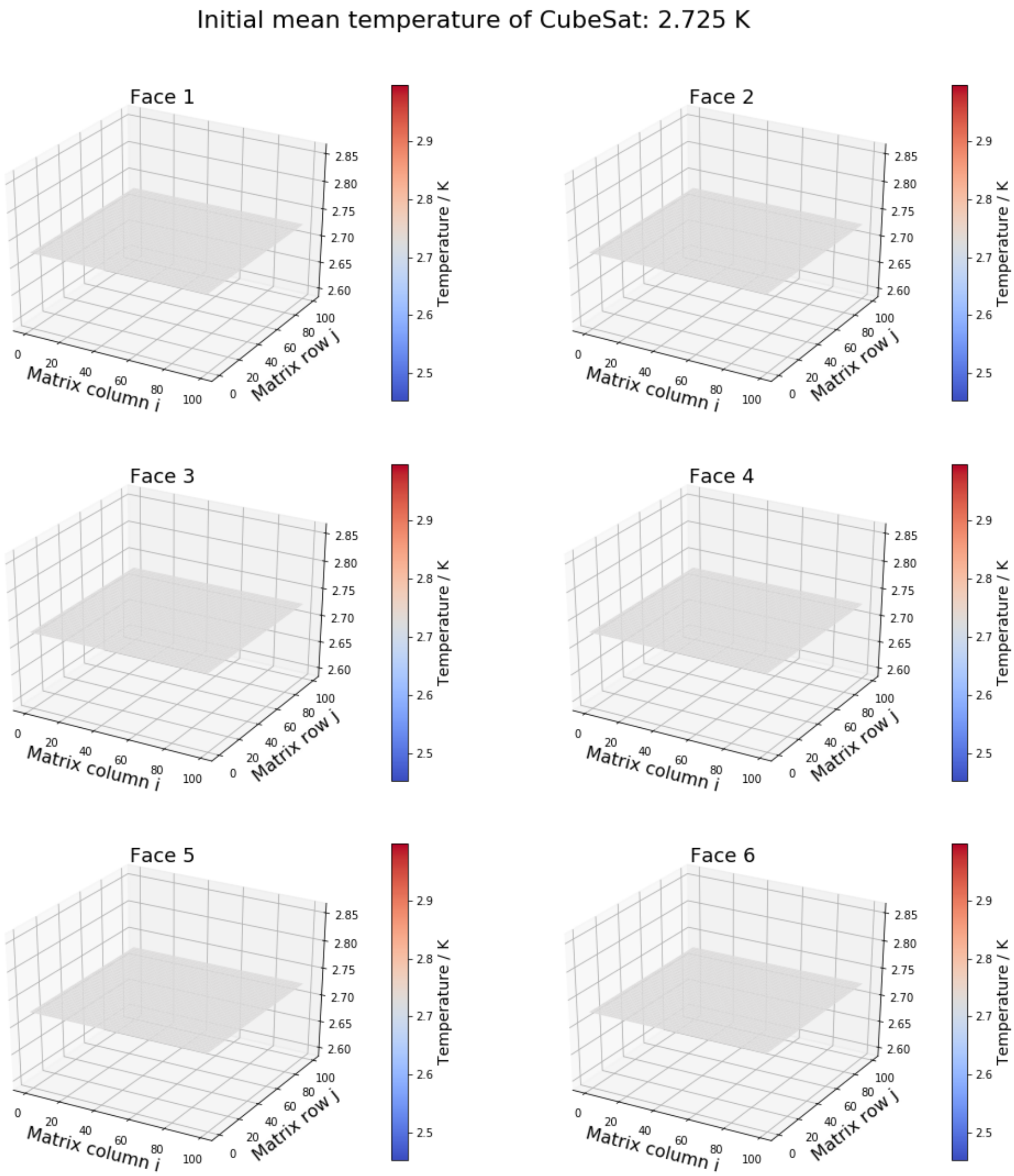



**Figure 4.1:** Initial temperature distribution of the CubeSat at $T = 0s$.

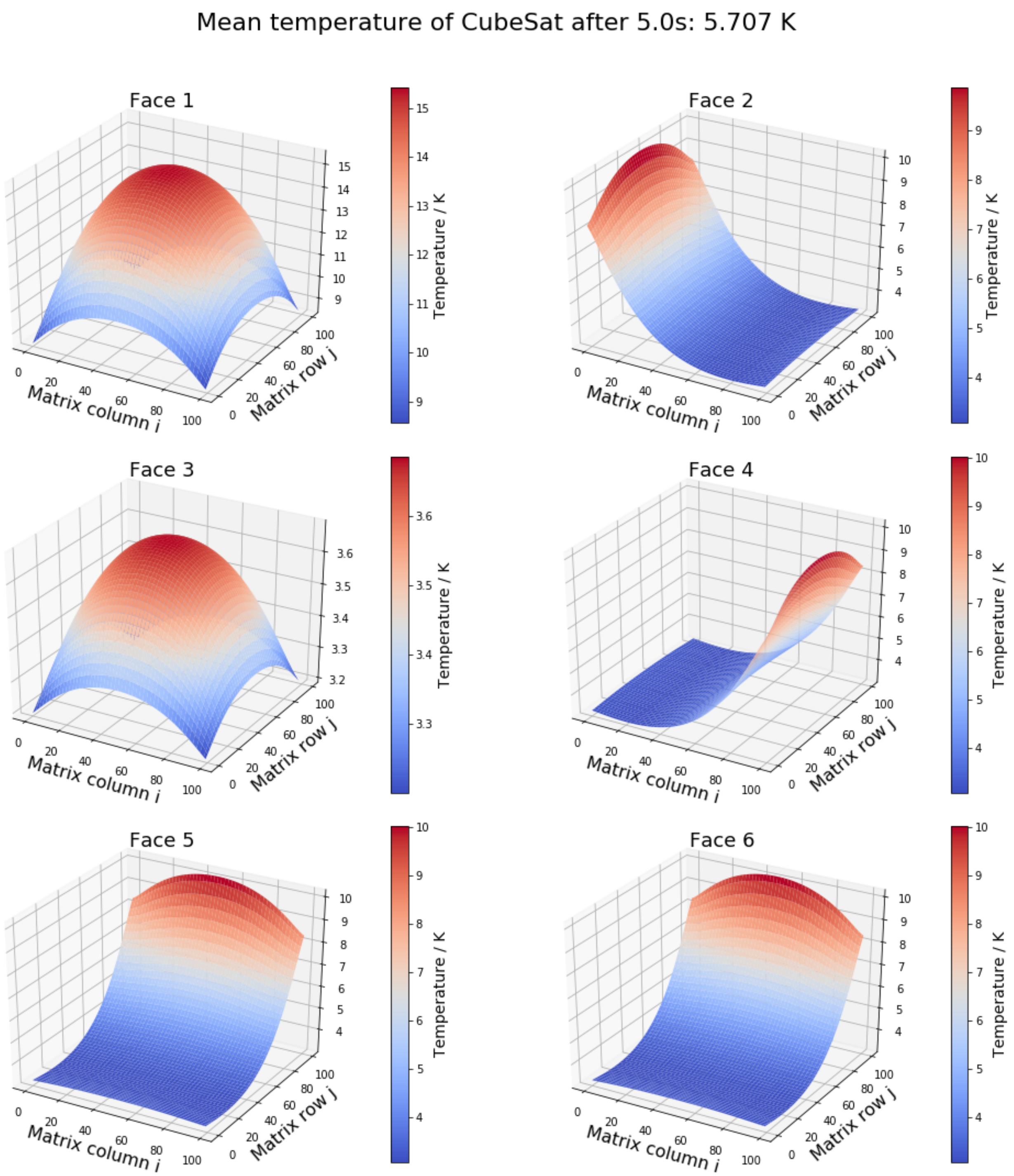



**Figure 4.2:** Temperature distribution of the CubeSat after $T = 5s$ ( 5000 timesteps).

## 4.2 Interpretation of Simulation Results

The simulation results in Figure 4.2 displays a few important features of heat evolution on the surface of the CubeSat under the constraints set out in Section 2. The interpretations of the result are as follows:

1. The simulation has successfully showed the most crucial attribute of heat diffusion where thermal energy flows from a hotter region into a colder region (i.e. thermal energy flows down the temperature gradient).
2. The symmetric setup of the CubeSat has once again resulted in the same temperature distribution across

Faces 2, 4, 5 and 6 at different orientations. Temperature distribution is symmetric along the $j$-axis of the matrix for Faces 2 and 4 whereas the distribution is symmetric in the $i$-axis of the matrix Faces 5 and 6. The different orientations are once again due to how the temperature matrix is defined in Figure 3.1.

3. Subsequently, temperature distributions on both Face 1 and Face 3 are also symmetric on their own since all four of their edges are attached to four symmetric faces. This result is particularly important confirms the isotropic distribution of heat in a homogeneous medium, or in other words, heat flows in the same way in all directions. This property and the direction of heat flow as stated in the first point validates the physicality of the simulation since the two crucial properties of homogeneous heat flow are obeyed.
4. The maximum temperature in Face 1 is higher than Face 3. Unsurprisingly, this means that the heat source that contributes most to the overall temperature increase of the CubeSat is the Sun.
5. The mean temperature of the CubeSat after $5s$ has doubled up compared to the initial temperature. This hints that the temperature of the CubeSat might continue to rise in the presence of the two heat sources - the Sun and the Earth. Since the energy radiation via the Stefan-Boltzmann Law has a $T^4$ temperature dependence, the system is unable to expel much the incoming heat when the temperature of the CubeSat surfaces are so low.
6. Also notice that the temperature is lower at the corners Faces 2, 4, 5 and 6 which are in direct contact with Face 1. This is because the same amount of heat is distributed across more element in the corners than the are in the centre of the edges.

Overall, the result of the simulation seems reasonable and makes physical sense. However, every numerical model has its limitations and this will be the following discussion.

## 4.3 Model Limitations

Recall that this model was developed using finite approximations to the 2-D heat equation which is then solved through the Explicit Euler method. Owing to the fact that stability criteria exists for the numerical method used, the length of time step $\Delta t$ has been constrained up to the millisecond scale. This implys that the model is not capable of exploring the evolution of temperature distribution across the surface of the CubeSat beyond a few seconds ($\sim 10,000$ time steps) as that would take up too much computational power.

# Section 5

# Conclusion

This project has demonstrated the applications of finite difference method on the 1-D and 2D heat equation. By discretising the 2-D heat equation, numerical techniques like the Euler Explicit Method can be emloyed to solve for the time evolution of temperature at a given point in physical space.

This numerical technique of solving the heat equation is then manoeuvred under some reasonable assumptions to develop a simple, preliminary model that described temperature distribution across the 2-D surface of a 1U CubeSat. The result of the simulation complies with the symmetry of the CubeSat and has shown two crucial feature of heat flow: one is that the direction of flow goes from a region of high energy into a region of lower energy, down the temperature gradient and the second feature is the isotropic flow of heat in a homogeneous medium. The results also showed that the mean temperature increases rather quickly in a short amount of time. Suffice to say, if the CubeSat were to be left in this setup for a long period of time, the temperature will increase boundlessly as the radiation of heat from the CubeSat via Stefan-Boltzmann Law is too slow to remove all the excess heat originating from the Sun and the Earth.

To sum it all up, although the model developed in this project still requires a lot of work before it can be put to use for a practical situation, it has shown physically valid results and hence is a good initialisation of the complex thermal modelling problem.

## 5.1 Future Measures

There are plenty of things that can be done to improve this heat evolutionary model, some of which are more complicated than others. Here are some feasible suggestions that can be taken as a next step to build on this simple model:

1. Based on the given model and under the same constraints, construct an improved model for the same hollow CubeSat where the surface now has finite thickness. From there, develop a model that is applicable to a solid 3-D 1U CubeSat using the 3-D heat equation. Note that the equation for conduction will now come into the picture due to finite thickness.
2. Start to consider orbital motions of the CubeSat around the Earth. This will mean that the incoming heat from the Sun is no longer continuous and will be 'switched off' temporarily when the line of sight of the CubeSat to the Sun is obstructed by the Earth as it goes around its orbit.
3. Incorporate a massless heat sink (cooling device) into the CubeSat to check if temperature of the CubeSat equilibrates at some point

# Appendix A

## 2D Heat Diffusion Algorithm in Python

The initial temperature of each central element of the 2-D plane can be put into a $100 \times 100$ matrix, `u0`. Then given a time step `ts`, the 2-D finite difference heat equation for the central elements can be solved using 'for-loop' that loops over the given number of time steps.

The boundaries (edges) of the plane are then set to be equal to the second outermost row or column. This is shown in the last 4 lines in the loop.

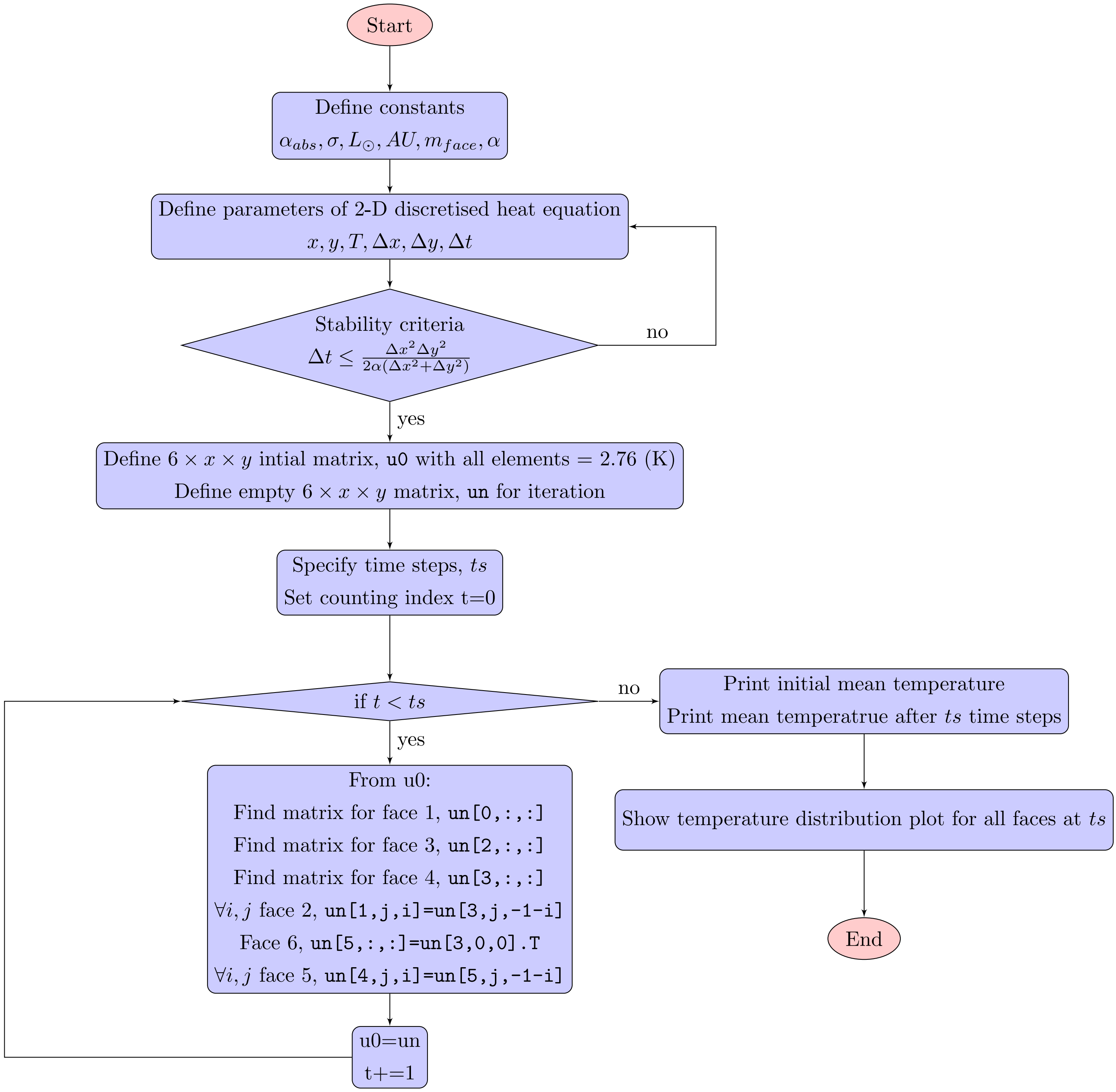
Code Starts

```
#ts = timestep
#u0 = initial temperature matrix for the given timestep
#un = temperature matrix after 'ts' iterations

un = u0.copy()

for i in range(ts+1):
   un[1:-1, 1:-1] = un[1:-1, 1:-1] + alpha * dt * (
          (un[2:, 1:-1] - 2*un[1:-1, 1:-1] + un[:-2, 1:-1])/dx2
          + (un[1:-1, 2:] - 2*un[1:-1, 1:-1] + un[1:-1, :-2])/dy2)
   un[0, :] = un[1, :]    #top edge
   un[-1, :] = un[-2, :]  #bottom edge
   un[:, 0] = un[:, 1]    #left edge
   un[:, -1] = un[:, -2]  #right edge
```

Code Ends

# Appendix B

## Flow Chart for the CubeSat Thermal Model



# References


[1] D. R. Williams (February 2018). Sun Fact Sheet. https://nssdc.gsfc.nasa.gov/planetary/factsheet/sunfact.html. Accessed: 2019-05-31.

[2] E. M. Greitzer, Z. S. Spakovszky, I. A. W. (2002). Kirchhoff's law and "real bodies". https://web.mit.edu/16.unified/www/FALL/thermodynamics/notes/node135.html. Section 19.2, Accessed: 2019-05-31.

[3] Fixsen, D. J. (2009). THE TEMPERATURE OF THE COSMIC MICROWAVE BACKGROUND. *The Astrophysical Journal*, 707(2):916–920.

[4] J. Wilson (August 2007). Thermal Diffusivity. http://www.electronics-cooling.com/2007/08/thermal-diffusivity/. Accessed: 2019-05-31.

[5] K. Funck, R. Nett, and A. Ostendorf (2014). Tailored beam shaping for laser spot joining of highly conductive thin foils. *Physics Procedia*, 56:750–758.

[6] L. Olsen-Kettle (2011). Numerical solution of partial differential equations. Chapter 2, Section 2.2, p15-16.

[7] Leon, N. D. (2001). Specific heat and heat capacity. http://www.iun.edu/~cpanhd/C101webnotes/matter-and-energy/specificheat.html. Accessed: 2019-05-31.

[8] M. Zeltkevic (April 1998). Forward and Backward Euler Methods. http://web.mit.edu/10.001/Web/Course_Notes/Differential_Equations_Notes/node3.html. Accessed: 2019-05-31.

[9] NASA CubeSat Launch Initiative (October 2017). CubeSat101 - Basic Concepts and Processes for First-Time CubeSat Developers. Section 1.1, p4.

[10] Paul Dawkins (April 2018). Section 9-1: The Heat Equation. http://tutorial.math.lamar.edu/Classes/DE/TheHeatEquation.aspx. Accessed: 2019-05-29.

[11] Serianni, A. (2014). Sustainability: Principles and practices. https://www3.nd.edu/~aseriann/SUSTAIN_Set_2.pdf. Accessed: 2019-05-31.